

solariNet

Developer & Architecture Reference — the implemented and planned system, end to end.

DOCUMENT

SolariNet_Developer_Reference

REVISION

1.0 · 2026-09-17

SOURCE BRANCH

stabilize/phase-0-containment

REPOSITORY

github.com/soren42/Solarinet

INTERNAL

Document control

This reference describes SolariNet — a self-hosted network monitoring, discovery, and control fabric written in standard C with a PHP dashboard — as it is actually built on the `akoria` homelab, alongside the architecture it is planned to become under the current stabilization program.

How this was assembled

The contents were reconstructed from ground truth, not memory: the committed source tree on branch `stabilize/phase-0-containment`; the rendered configuration live on the primary host `xenon`; the DNS source-of-truth `netdb/akoria-hosts.yml`; and an authorized live discovery pass across the homelab (host-liveness sweep plus read-only SSH fact-gathering). Network discovery on this infrastructure is routine, authorized administration of the operator's own network.

Reading the provenance flags

Every non-trivial claim is marked so a reader can tell measured fact from reasoning:

OBSERVED Read directly from source code, live host state, or a discovery scan during assembly.

INFERRED Deduced from surrounding evidence; not directly confirmed. Treat as a lead, not a guarantee.

Capability maturity is shown with the same labels the dashboard uses (planned track I0):

PROTOTYPE **LAB** **PRODUCTION** **AUTHORITATIVE**

A feature marked **LAB** works but is not trusted for unattended production use; several such capabilities are explicitly frozen (see §10).

SCOPE & HONESTY NOTE

SolariNet is under a declared **feature freeze**. Its own architecture review found that feature breadth had outrun enforcement and durability. This document therefore separates what is implemented and load-bearing from what exists in code but is not yet trustworthy — the distinction is the point, not a footnote.

Conventions

- Hostnames, IP/MAC addresses, ports, node IDs, file paths, verbs, and code are set in IBM Plex Mono — the data face.
- Hosts are named for periodic-table elements and compounds (an RFC 2100 convention); the name carries no functional meaning. Function is expressed as a DNS CNAME.
- The control-plane binaries are `solariClient`, `solariMonitor`, `solariServer`; the wire protocol is SCP (SolariNet Control Protocol).

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System overview

SolariNet is a self-hosted fabric that watches a homelab and acts on it: edge agents report host health, a monitor tier probes reachability, and a central server persists state, raises alerts, drives discovery, and brokers operator commands — all in standard C, fronted by a build-step-free PHP dashboard.

What it is

The system is a clean **three-tier control plane** — `solariClient` (per-host agent), `solariMonitor` (on-the-wire prober), and `solariServer` (ingest, storage, alerting, control) — speaking a single custom binary protocol, SCP, over mutually-authenticated TLS. Around that core sit a MariaDB datastore, a RabbitMQ notification bus, a DNS/AD System of Record with change-data-capture, and an identity stack (Keycloak federated to Samba Active Directory). The operator sees all of it through one dashboard served by Apache and PHP-FPM.

The codebase holds to two hard house rules that shape every decision below: the application is **standard C with associated Makefiles**, and the dashboard is **PHP with no build step** — React and Babel are vendored and JSX is transpiled in the browser at load. There is no Node.js in the runtime, no bundler, and no server-side JavaScript anywhere. **OBSERVED** (no `package.json`, `composer.json`, or `lockfile` exists in the tree).

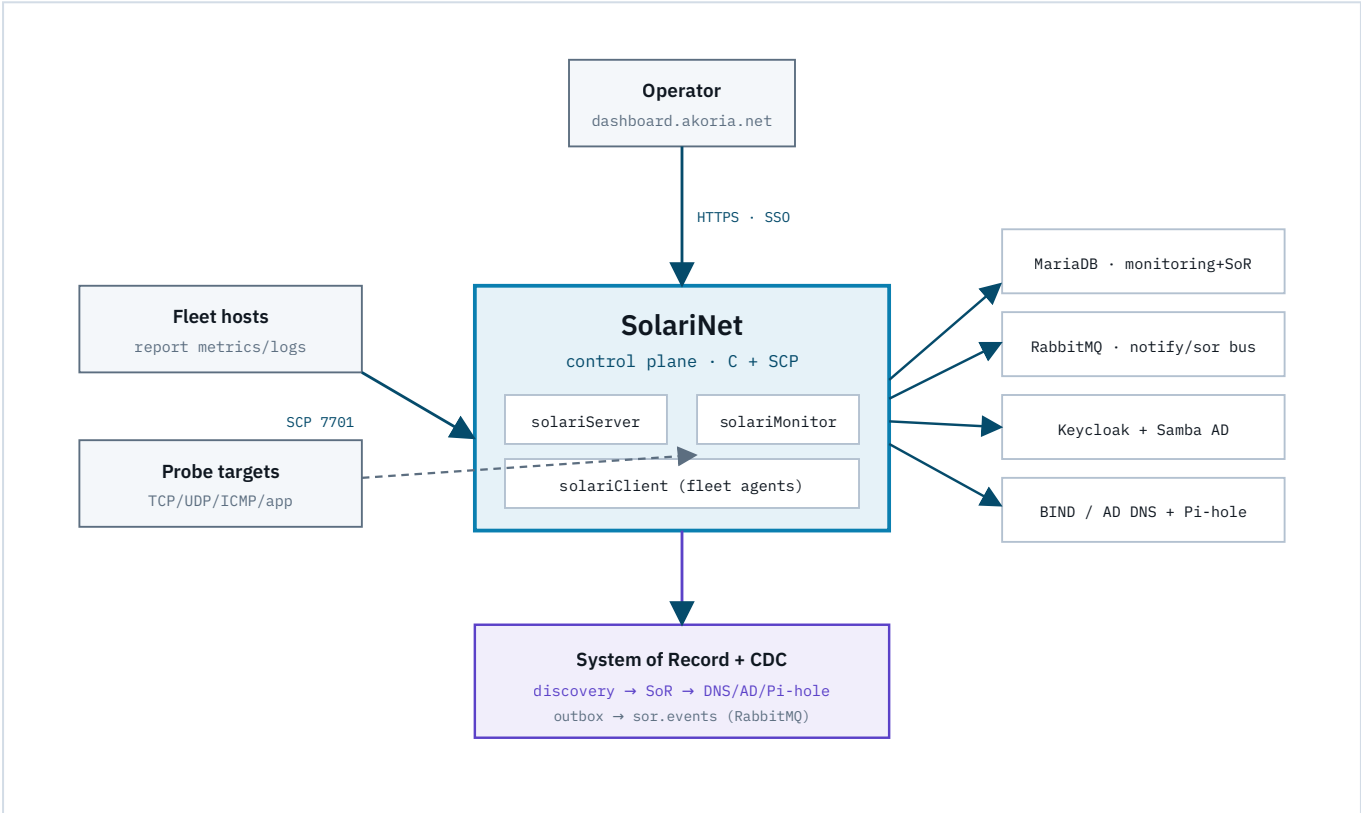


Figure 1.1 – System context. Solarinet's C control plane at center; operators, fleet, and probe targets around it; datastore, bus, identity, and DNS as backing services. Provenance: **OBSERVED** from source + live config.

The honest caveat: a fleet-of-one in practice

READ THIS BEFORE TRUSTING THE TOPOLOGY

The architecture is a genuine three-tier split, and the monitor tier's rendezvous-hash ownership is implemented in code. **In production, however, nearly every node runs the client role only**; the monitor tier and its k-of-n redundancy are not yet load-bearing. Treat multi-monitor resilience as designed-and-coded, not proven. Section 11 tracks the work (F-19/F-20) that is meant to make the claim true.

Technology stack

CORE TECHNOLOGY CHOICES – FROM DOCS/ARCHITECTURE.MD

LAYER	CHOICE	WHY (PER RATIONALE DOC)	FLAG
Control-plane language	ISO C99	Portable, dependency-light, functions-only; opaque-handle pattern, no OOP emulation.	OBS
Wire transport	nng + mbedTLS	Scalable-protocol sockets over mutually-authenticated TLS (tls+tcp://).	OBS
Wire protocol	SCP	Length-prefixed 32-byte big-endian header + TLV payload + CRC-32; golden-frame fixtures immutable per version.	OBS
Edge spool	SQLite	Durable client-side store-and-forward when the server is unreachable.	OBS

LAYER	CHOICE	WHY (PER RATIONALE DOC)	FLAG
Datastore	MariaDB / InnoDB	Monitoring DB + System of Record; all timestamps stored UTC.	OBS
Message bus	RabbitMQ	Durable topic exchanges for notifications and SoR change events; Mosquitto for device MQTT.	OBS
Dashboard	PHP 8.x + Apache	No framework, no Composer, no build step; React/Babel vendored, JSX transpiled in-browser.	OBS
Identity	Keycloak (OIDC)	Confidential-client auth-code flow, federated to Samba AD over LDAPS.	OBS
Support daemons	Python (deploy/only)	Discovery, notify, sorsync, unifi, etc. — never inside the C/PHP tiers.	OBS

Two domains, one fabric

DNS is split by purpose. `akoria.net` is the general forward zone, authoritative on xenon (BIND primary) with secondaries on radium and steel. `akoria.org` is the Active-Directory / Kerberos realm, authoritative on radium (Samba DC, BIND9_DLZ), mirrored hourly from `.net`. Hosts get an element name as their A record; functions are layered on as CNAMEs (dashboard, sso, git, db, ...). INFERRED: `solarian.net` appears only as branding — no zone for it exists in the repo.

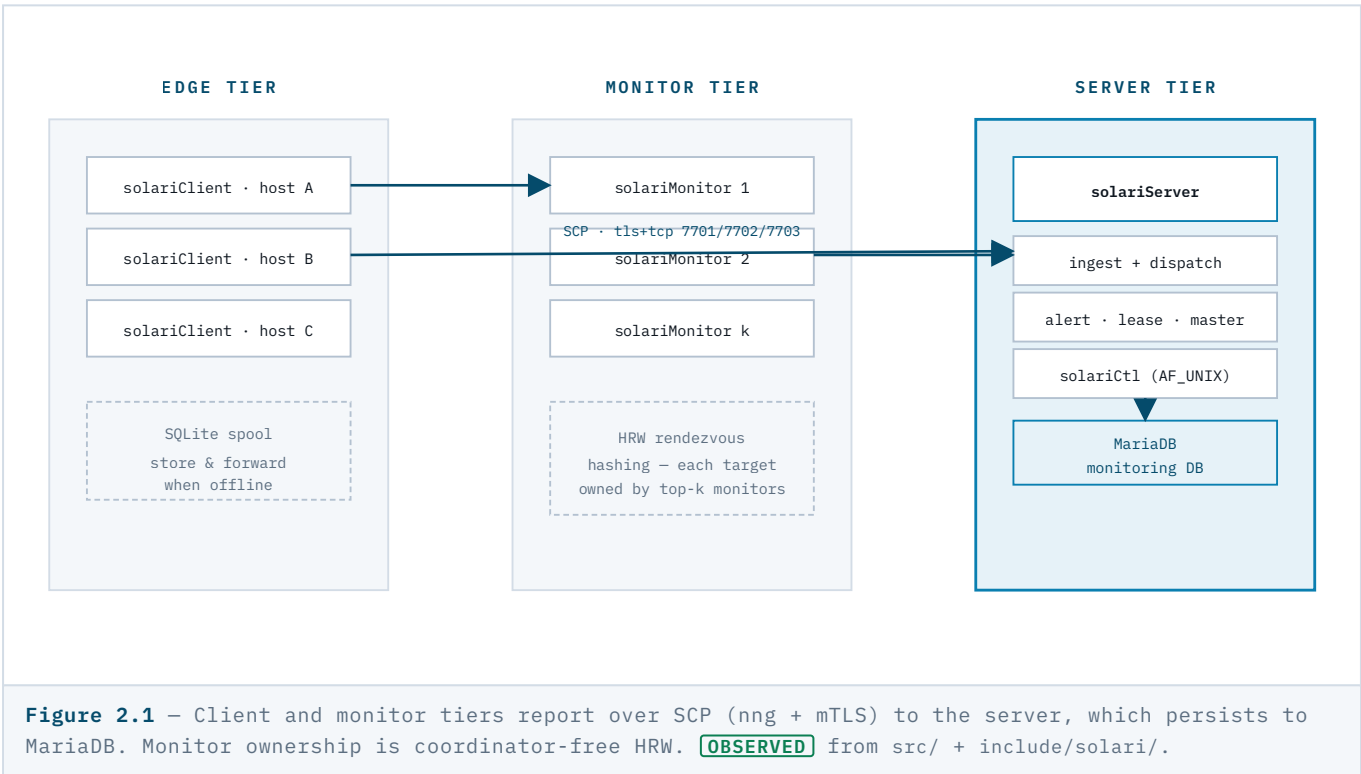
Control-plane architecture

Three C programs, one shared static library, and one binary protocol. Everything the fabric knows flows through `solariServer`; everything an operator does to it flows through a single local socket into the same process.

2.1 Three-tier topology & SCP transport

All three daemons link one static library, `libsolari` (`lib/solari/`), which owns framing, TLV, crypto, config, JSON, logging, time, the SQLite spool, and the `nng` transport. The core is zero-dependency; the transport (`solariNet.c`), spool (`solariSpool.c`), and reporter (`solariReporter.c`) compile only when their feature flags (`SOLARI_WITH_IO`, `SOLARI_WITH_SQLITE`) are set.

- solariClient** Per-host agent. Collects host metrics, log tails, and process info behind a platform-abstraction layer (only `platLinux.c` exists today); reports over SCP; spools to SQLite when offline. Discovery advertises via mDNS/avahi.
- solariMonitor** On-the-wire prober (`probeNet.c`: ICMP via unprivileged `SOCK_DGRAM` with fallback, TCP-connect, UDP, app-level). Targets are split across the monitor fleet by **rendezvous (HRW) hashing** at a replication factor — no coordinator. Linux-only.
- solariServer** The hub: validates and dispatches frames, persists to MariaDB, runs the alert engine, discovery enrichment, SNMP polling, provisioning, the leadership lease, and hosts the `solariCtl` operator bridge. Also builds `solariSnmPoll` and `solariCtlClient` from the same core.



SCP frame & ingest gate

Every frame is validated by `solariFrameParse` (magic, version, CRC) before dispatch. The ingest path then derives the presenting node's **role from its TLS certificate CN** (`<role>.<fqdn>`), rejects any message type illegal for that role (`ERR_AUTH_ROLE`), and de-duplicates by (`sourceNodeId`, `seqNo`) via a process-local ring — best-effort, explicitly not durable exactly-once. Dispatch routes by message type:

SCP MESSAGE DISPATCH — SRC/SERVER/SERVERINGEST.C

MESSAGE	ROUTED TO	EFFECT
CLIENT_REPORT	serverDb + serverAlert	Host metrics persisted; host-scope alert rules evaluated
MONITOR_REPORT	serverDb + serverMaster + alerts	Probe results persisted; ownership reconciled
WATCHDOG	serverDbTouchNode	Liveness heartbeat
DISCOVERY_ADVERT	serverDiscovery	Discovered-host table updated
TOPOLOGY_REPORT	serverTopology	LLDP/topology edges
SURVEY_RESP / CONTROL_RESULT	serverControl	Command convergence tracking

2.2 solariServer internals & the solariCtl bridge

`solariCtl` is **not a separate binary** — it is a module compiled into `solariServer` (`src/server/solariCtl.c`) that listens on an `AF_UNIX` socket (default `/run/solari/solariCtl.sock`). It is the only way the dashboard mutates state. The protocol is line-oriented text:

```
# request / reply grammar (solariCtl.c)
request : VERB [k=v ...]\n
reply   : OK [k=v ...]\n | ERR <code> <message>\n
```

Destructive verbs (marked `*` in the verb table) require an `op=<operator>` attribution; `DECOMMISSION` additionally requires a two-step `confirm=<token>` handshake — the first call returns a token without acting. The same module holds the internal CA/CSR signing material (mbedtls x509), so **the PHP dashboard never holds a server certificate or CA key**.

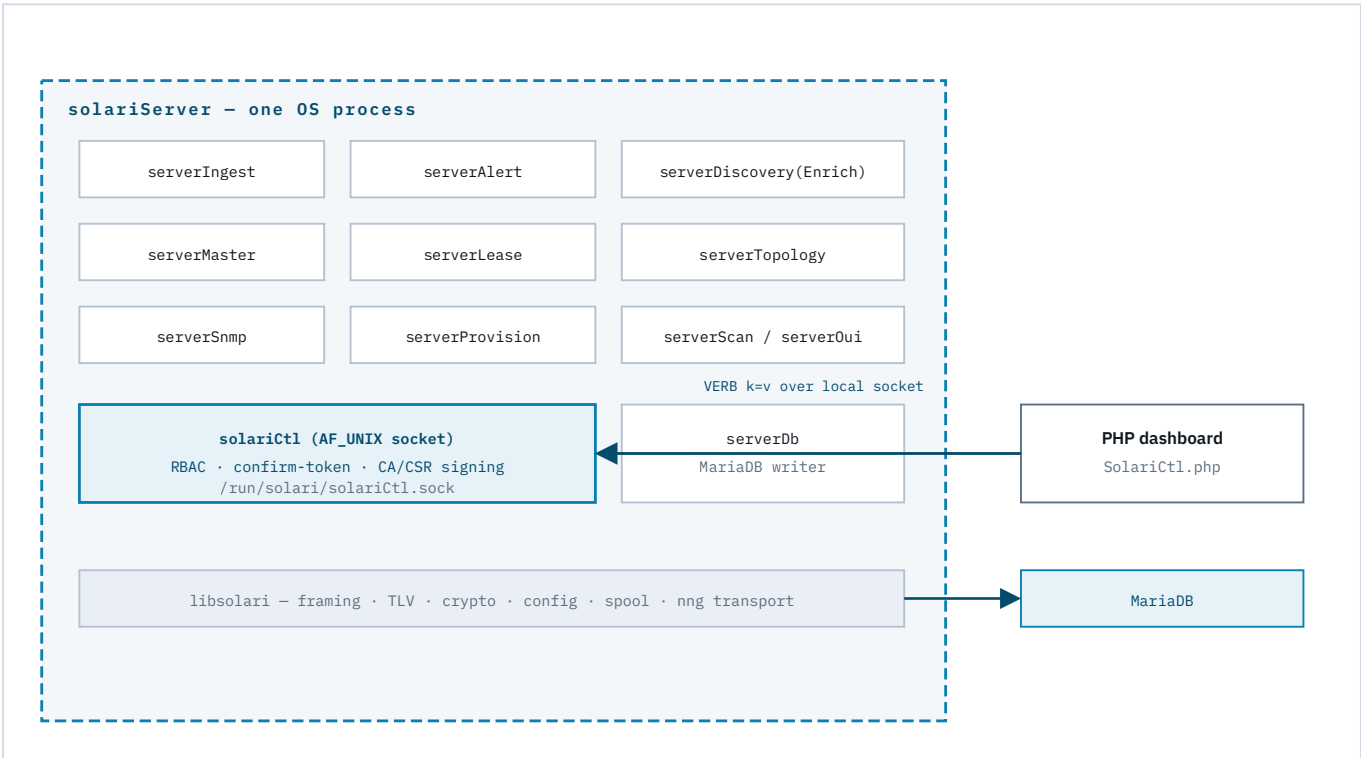


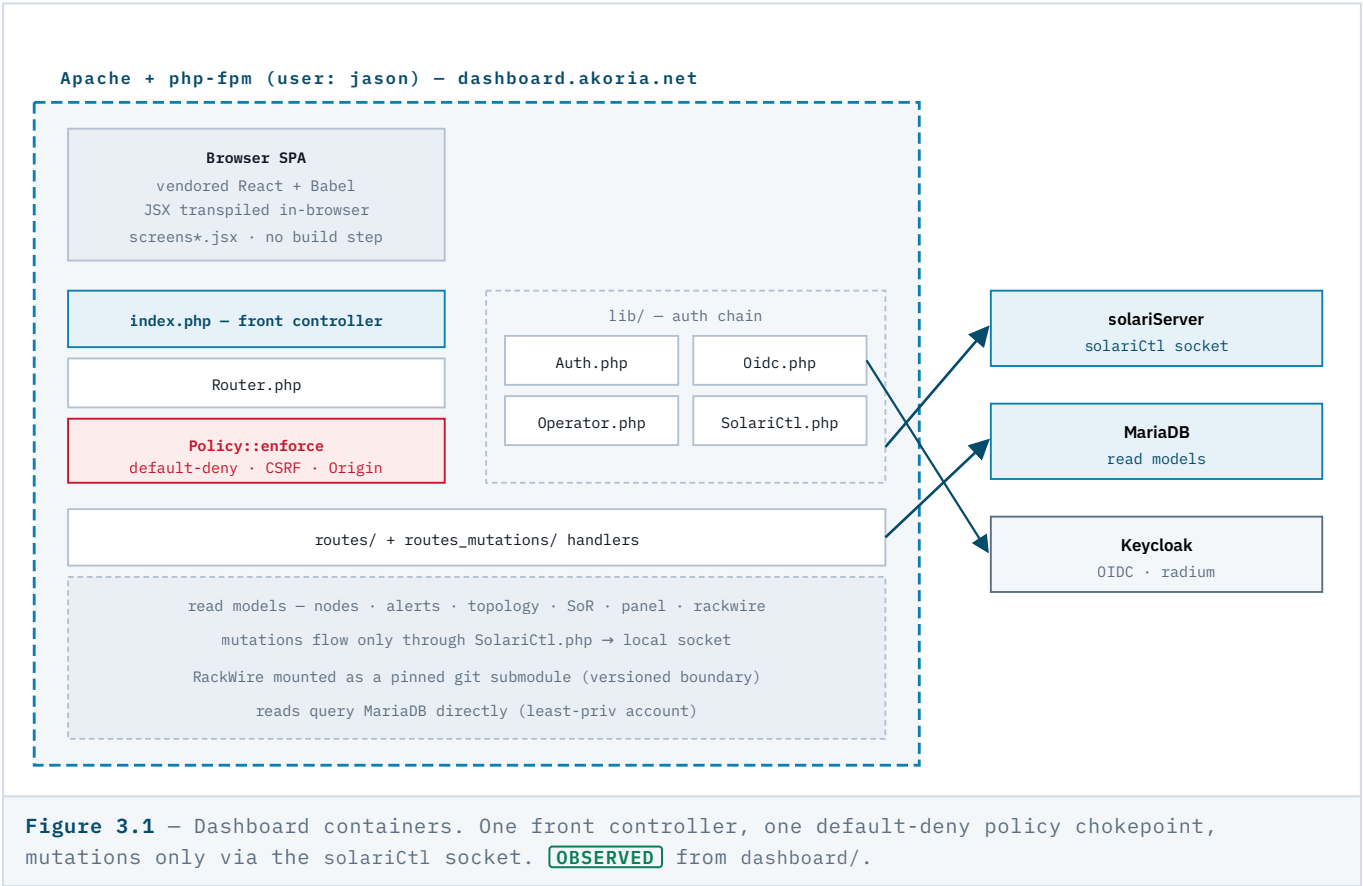
Figure 2.2 – Inside solarisServer: the operator bridge solarisCtl and the CA material live in-process; the dashboard reaches them only through the local socket. **OBSERVED** from src/server/.

Subsystem container views

Four subsystems carry most of the system's behaviour outside the C core: the PHP dashboard, the notification pipeline, the SoR change-data-capture path, and the discovery/enrichment loop. Each is drawn here as a nested container view — process boundary, the containers inside it, and the trust edge each one crosses.

3.1 The PHP dashboard

The dashboard is a single Apache + PHP-FPM application with **one front controller** (dashboard/api/index.php) and a custom Router. Every request passes one chokepoint — Policy::enforce — which is **default-deny**: a route with no explicit grant is refused, and every non-GET additionally clears a CSRF synchronizer-token + Origin check before the handler runs (Phase-0 hardening, commits 2d74adf / 1812a9e). The SPA is vendored React + Babel-standalone; **JSX is transpiled in the browser** at load. There is no bundler and no package.json. php-fpm runs as the unprivileged user jason and reaches the C core only through the solariCtl UNIX socket — it never holds a certificate or a DB super-grant.



3.2 Notification & SoR-CDC pipelines

Two Python daemons in `deploy/` bridge the fabric to the outside world over RabbitMQ. Neither is inside the C or PHP tiers — both are support glue, and both declare their exchanges defensively (there is no canonical broker config committed to the repo, a gap tracked below).

notify pipeline `solariServer`'s alert engine publishes to the durable topic exchange `notify.events` (vhost `solari`, `amqp://...@10.5.2.50:5672/solari`). `notify.service` / `alertbridge.service` consume and fan out to delivery channels (Apple push bridge, etc.).

SoR CDC Writes to the SoR/CMDB DB fire `sor_outbox` triggers; `sor-emit` drains the outbox to the `sor.events` exchange; `sor-apply-dns` and `sor-apply-pihole` consume and reconcile DNS/Pi-hole. The reverse path (`sor-reconcile`) folds discovery back into the SoR.

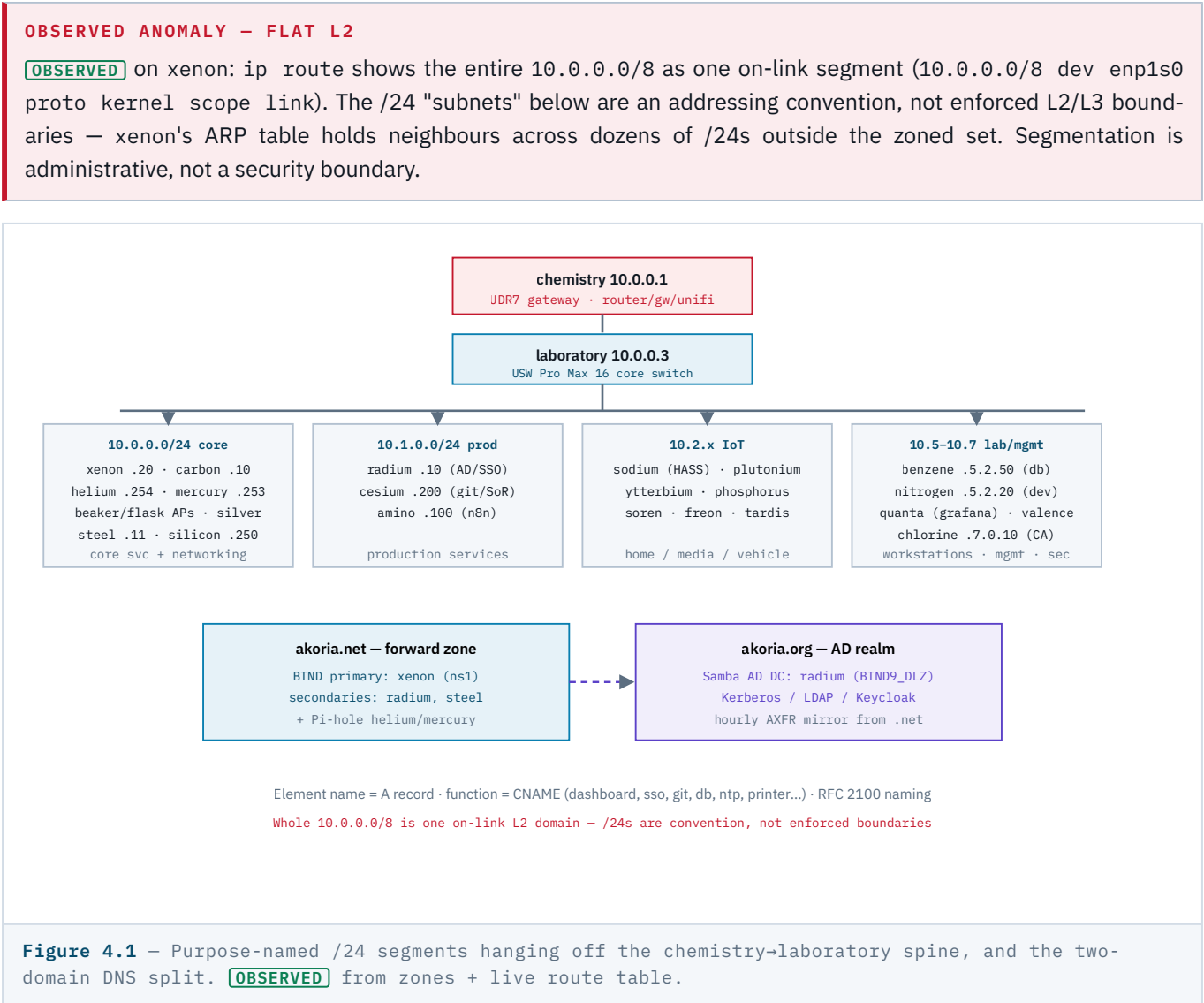
KNOWN GAP — F-08

The CDC path can **permanently lose events** as built: publishes are not confirm-gated and consumers ack before applying. Phase-2 of the stabilization program adopts publisher-confirms, ack-after-apply, and a reconcile sweep. Until then, treat DNS/Pi-hole convergence from the SoR as eventually-consistent-at-best, not guaranteed.

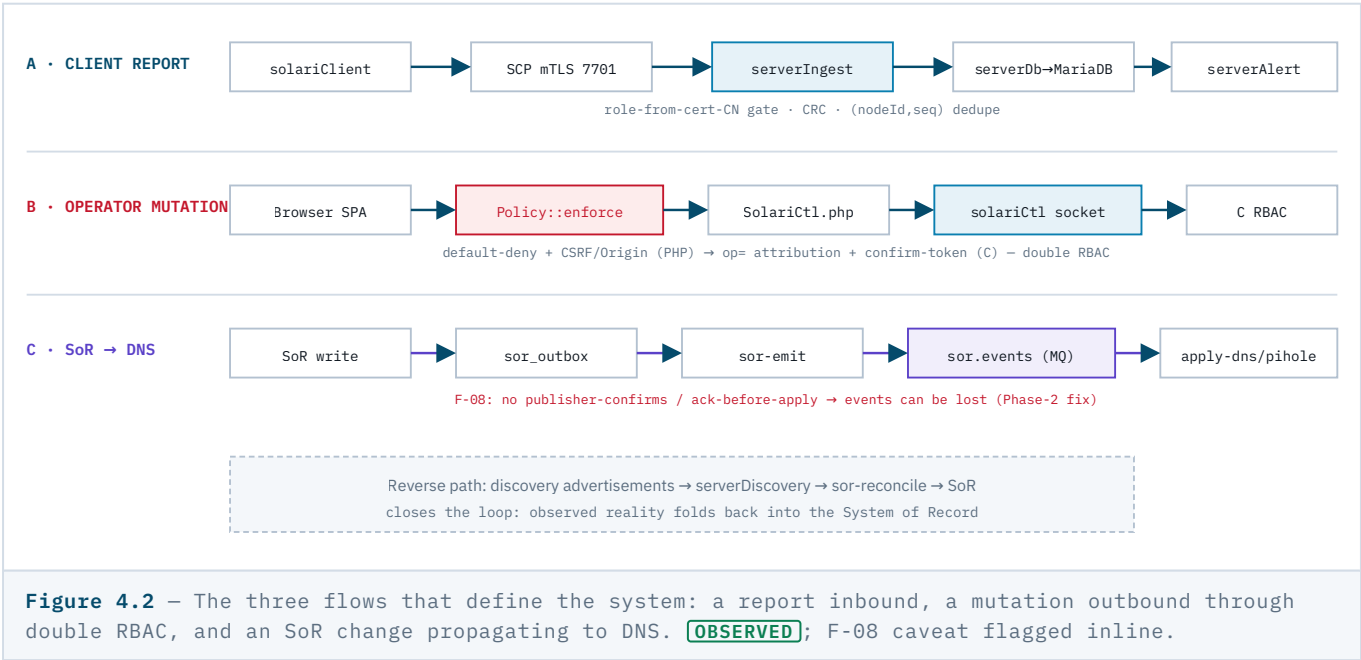
Network topology & data flows

The fabric rides a flat 10.0.0.0/8 L2 domain segmented by convention into purpose-named /24s. This section maps the segments, then traces the three flows that matter: a client report, an operator mutation, and an SoR change propagating to DNS.

4.1 Network segmentation



4.2 Three canonical flows



System inventory

Every host on the fabric, by name, manufacturer, type, and network address. Rows fuse the committed System of Record (`netdb/akoria-hosts.yml`) with a live discovery sweep from xenon on 2026-09-17. Each row is flagged **OBS** where a fact was observed live and **SoT** where it comes only from the committed record.

AUTHORIZATION & METHOD

The discovery below is ordinary network administration on infrastructure the operator owns. Method: `nmap -sn -PR` host discovery across the 16 zoned /24s, offline IEEE OUI lookup against the local `nmap-mac-prefixes` table, and read-only SSH fact-gathering (`hostnamectl / dmidecode`) where a key was accepted. **No service/OS scan and no SNMP** completed — port/OS columns are honestly blank, not empty by absence of service. MACs are only shown where captured in the live ARP snapshot; none are synthesized.

5.1 Subnets

ADDRESSING SEGMENTS — 10.0.0.0/8 (ONE FLAT L2 DOMAIN; /24S ARE CONVENTION)

CIDR	PURPOSE	LIVE HOSTS UP	FLAG
10.0.0.0/24	Core networking + core services	chemistry, laboratory, carbon, xenon, beaker, flask, silver, catalyst, accelerant (+ 5 un-named)	OBS
10.0.1.0/24	Workstation + recovered/legacy (boson, muon, gallifrey, hydrogen)	none up at scan	OBS
10.1.0.0/24	Production — AD/DC, Forgejo/SoR, n8n	radium, amino, cesium	OBS
10.2.0.0/24	IoT — Home Assistant, voice	sodium, plutonium	OBS
10.2.2.0/24	IoT — media / console	ytterbium, phosphorus	OBS
10.2.3.0/24	IoT — appliance, water dish	soren	OBS
10.2.4.0/24	IoT — thermostat	freon	OBS
10.3.0.0/24	Peripherals — radio, scanner, UV printer	none up at scan	OBS
10.4.1.0/24	Personal — tablet (callisto)	none up at scan	OBS
10.4.4.0/24	Personal — vehicle (tardis)	tardis	OBS
10.5.0.0/24	Workstations/SBC — tungsten	none up at scan	OBS
10.5.2.0/24	Workstations/SBC — nitrogen, benzene, astatine	nitrogen, benzene	OBS
10.5.87.0/24	Lab gear — eleksmaker laser/CNC	none up at scan	OBS
10.6.0.0/24	Network mgmt — Grafana, control panel, outpost	valence	OBS

CIDR	PURPOSE	LIVE HOSTS UP	FLAG
10.6.6.0/24	Network mgmt — GPS/PTP time source	none up at scan	OBS
10.7.0.0/24	Security — NVR hub, internal CA	photon, chlorine	OBS
203.0.113.0/24	RFC 5737 documentation range — host disc is a placeholder, not real infra	not scanned	SoT

Reverse zones 10.1.4/24, 10.2.5/24, 10.3.2/24, 10.4.2/24 are declared in BIND but hold no hosts in the current SoT — stale zones or an incomplete host list. OBS

5.2 Host inventory — infrastructure & services

CORE INFRASTRUCTURE HOSTS. MAC = LIVE ARP CAPTURE; HARDWARE CONFIRMED BY SSH WHERE NOTED.

HOST	IP	MANU-FACTUR-ER	TYPE / MODEL	ROLE · KEY CNAMES	MAC (OBS)	FLAG
chem-istry	10.0.0.1	Ubiqui-ti	UDR7 Dream Router 7	gateway/firewall/UniFi · router,gw,unifi,ns0	74:fa:29:41:f0:57	OBS↑
labo-ratory	10.0.0.3	Ubiqui-ti	USW Pro Max 16	core switch (etherlight-ing)	9c:05:d6:e4:f1:ac	OBS↑
beaker	10.0.0.7	Ubiqui-ti	U7 Pro	WiFi 7 AP	a8:9c:6c:4c:9a:d0	OBS↑
flask	10.0.0.8	Ubiqui-ti	U7 Pro	WiFi 7 AP	a8:9c:6c:6a:09:c0	OBS↑
xenon	10.0.0.20	youyee-too	X1 SBC · Celeron N5095 · 16 GiB · Ubuntu 26.04	SolariNet serv-er+dashboard+BIND prima-ry · dash-board,monitor,solari,ns1	(self)	OBS✓ssh
carbon	10.0.0.10	UGREEN	DXP4800 Plus · Debian 12 · box-hostname "NAS-X"	NAS · nas,nas-x,c	6c:1f:f7:0c:76:0f	OBS✓ssh
helium	10.0.0.254	Rasp-berry Pi	Raspberry Pi (model unspec.)	Pi-hole primary/resolver · ns2,hole,dns1,he	—	SoT
mer-cury	10.0.0.253	Ice-Whale	ZimaBoard	Pi-hole secondary · ns3,hole2,dns2	—	SoT
steel	10.0.0.11	Ice-Whale	ZimaBlade	akoria.net secondary NS (⚠ SoT still names it iron /db — drift)	—	SoT
sili-con	10.0.0.250	Rasp-berry Pi	Raspberry Pi 4	edge proxy (Caddy auto-ACME)	—	SoT

HOST	IP	MANUFACTURER	TYPE / MODEL	ROLE · KEY CNAMES	MAC (OBS)	FLAG
scandium	10.0.0.12	Ice-Whale	ZimaBoard	services (n8n/Uptime-Kuma) – offline/planned	–	SoT
halogen	10.0.0.2	Ubiquiti	UGC Max	old primary router (offline, minirack pending)	–	SoT
radium	10.1.0.10	Raspberry Pi	CM5 · Debian 13 · arm64	Samba AD DC + Keycloak + BIND9_DLZ/.org · ad,sso,ldap,kdc	2c:cf:67:b5:ab:7d	OBS✓ssh
cesium	10.1.0.200	Dell	PowerEdge R410 · openSUSE Tumbleweed	Forgejo + SoR/CMDB MariaDB · git,ad-backup	d4:ae:52:97:44:e9	OBS✓ssh
amino	10.1.0.100	Ice-Whale	ZimaBoard 832 · ZimaOS · Celeron N3450 · 8 GiB	n8n workflow automation · n8n	00:e0:4c:17:7c:97	OBS✓ssh
benzene	10.5.2.50	HP	EliteDesk 800 G6 Mini · i5-10500T · openSUSE	MariaDB (SoR mirror) + fleet netboot :8080 · db	e0:70:ea:b0:13:e6	OBS✓ssh
nitrogen	10.5.2.20	Raspberry Pi	Pi 500 · Debian 13 · box- hostname "dev-family-pi01"	dev / IDE · dev	2c:cf:67:bc:d5:2e	OBS✓ssh
tungsten	10.5.0.5	Minisforum	MS-R1	AI/dev workstation (AD-joined)	–	SoT
astatine	10.5.2.10	Argon41	OneUp SBC	SBC workstation · laptop	–	SoT
quanta	10.6.0.10	Raspberry Pi	Pi 5	Grafana · grafana	–	SoT
valence	10.6.0.20	FriendlyElec	NanoPi-R1 (armhf)	monitoring outpost	02:81:12:d7:eb:db (locally-admin MAC)	OBS↑
chlorine	10.7.0.10	Khadas	VIM4 · Ubuntu 22.04 · arm64	internal CA/key server (step-ca) · ca,acme,pki	c8:63:14:72:87:81	OBS✓ssh
tachyon	10.6.6.10	Particle	Tachyon	GPS/PTP time source · ntp	–	SoT

OBS↑ = up + MAC observed live; **OBS/ssh** = hardware confirmed by SSH fact-gather; **SoT** = committed record only, host down or unreachable at scan.

5.3 Host inventory – endpoints, IoT & peripherals

CLIENT / IOT / PERIPHERAL HOSTS (LARGELY SOT; MAC WHERE LIVE-OBSERVED).

HOST	IP	MANU-FACTUR-ER	TYPE / MODEL	ROLE	MAC (OBS)	FLAG
silver	10.0.0.85	Epson	ET-8550	printer · printer	64:c6:d2:5b:c3:2a	OBS↑
catalyst	10.0.0.90	Hitron	MoCA adapter	Family Room MoCA bridge	dc:36:0c:d5:db:7d	OBS↑
accelerant	10.0.0.91	Hitron	MoCA adapter (inferred from OUI)	unknown ("recovered from DNS")	dc:36:0c:d6:11:b5	OBS↑
hydrogen	10.0.1.50	Apple	Mac Mini M4 Pro	primary workstation	—	SoT
sodium	10.2.0.20	—	unknown	Home Assistant · hass	88:a2:9e:24:d3:dc (Raspberry Pi OUI)	OBS↑
plutonium	10.2.0.100	Amazon	Echo Dot	voice assistant	f8:54:b8:d2:ac:76	OBS↑
ytterbium	10.2.2.2	Sony	PlayStation 5	media / console	78:c8:81:c1:04:b6	OBS↑
phosphorus	10.2.2.10	Google	Google TV 4K	media (Family Room)	b8:7b:d4:ff:ce:6d	OBS↑
soren	10.2.3.10	Espressif (OUI)	LR4 MBR appliance	MBR appliance	c4:d8:d5:6e:32:c0	OBS↑
dihydroxide	10.2.3.20	—	Pawaii	smart water dish	—	SoT
freon	10.2.4.10	Google/Nest	Nest thermostat	thermostat	3c:31:74:dd:3f:87	OBS↑
tardis	10.4.4.10	Tesla	Model 3	vehicle	cc:88:26:26:f3:db (LG Innotek OUI)	OBS↑
calisto	10.4.1.10	Apple	iPad Pro 12.9	tablet	—	SoT
photon	10.7.0.1	Eufy	Eufybase NVR	NVR hub	04:17:b6:88:66:75	OBS↑
bromide	10.3.0.80	Epson	RR-600W	scanner · scanner	—	SoT

HOST	IP	MANU-FACTUR-ER	TYPE / MODEL	ROLE	MAC (OBS)	FLAG
plat-inum	10.3.0.100	Eufy-Make	E1 UV printer	UV print-er	—	SoT
radio	10.3.0.34	—	"Radio Pi"	radio backend	—	SoT
eleks-maker	10.5.87.15	Eleks-Maker	laser/CNC con-troller	lab tool	—	SoT
bo-son / muon	10.0.1.125/.126	—	ArkKVM	KVM units · arkkvm1/2	—	SoT
lithi-um	10.6.4.33 / 10.6.3.33	Arduino	Imola · Debian 13	status-panel host ▲ on unzoned subnets	—	OBS✓ssh

RECONCILIATION QUEUE — OBSERVED DRIFT & UNKNOWNNS

- **Five un-named live hosts** answered ARP in 10.0.0.0/24 with no PTR and no SoT record: 10.0.0.6, .31, .32, .33, .160. No MAC captured (cache aged out). Candidate unlabeled or rogue devices — worth identifying. OBS
- **lithium** sits on 10.6.3.0/24 and 10.6.4.0/24, neither of which is a zoned subnet — a real SoT host on unad-vertised address space. OBS
- **iron → steel** rename: the live zone serves steel as the DNS secondary, but the SoT reserved: block still calls the box iron with a different (db) role. Same hardware, unreconciled record. OBS
- **nitrogen** DNS name vs. box hostname dev-family-pi01; SoT lists it as a Pi 500, live kernel is a CM5-class 2712 — confirm the exact model. OBS
- **"Recovered from DNS"** hosts (quartz, tin, accelerant, bunsen, gallifrey, strangeness) have genuinely unknown hardware — reconstructed from production DNS, never deliberately assigned. SoT

Component reference

The runnable parts of the system — daemons, support services, and the ports and sockets that connect them.

6.1 Binaries (C tier)

COMPILED FROM THE SINGLE CMAKE C99 TREE; ALL LINK LIBSOLARI.

BINARY	RUNS ON	LISTENS / CONNECTS	PURPOSE
solariServer	xenon	SCP tls+tcp 7701–7703 in; solar-ictrl.sock; MariaDB out	Ingest, persist, alert, discovery, SNMP, lease/master, operator bridge, CA signing
solariClient	fleet hosts	SCP out to server; SQLite spool local; mDNS advertise	Per-host metrics/logs/process reporting with offline store-and-forward
solariMonitor	designated hosts	SCP out; ICMP/TCP/UDP/app probes to targets	On-the-wire reachability probing; HRW-partitioned target ownership
solariSnmpPoll	xenon	SNMP out; SCP in-process	SNMP metric collection (built from the same core)
solariCtlClient	any	solariCtrl.sock	CLI to the operator bridge for scripts/tests

6.2 Services & systemd units

DEPLOYED UNITS — DEPLOY/HOST-CONFIGS/XENON/*.SERVICE AND THE WIDER FABRIC.

SERVICE	HOST	ENDPOINT	NOTES / FLAG
SolariNet dashboard	xenon	dashboard.akoria.net (HTTPS vhost, :9443 intranet)	Apache + php-fpm (user jason) OBS
solarinet-server / -monitor / -snmp	xenon	systemd units	C daemons; hardened units OBS
notify / alertbridge	xenon	consume notify.events	alert fan-out to delivery bridges OBS
sor-emit / -apply-dns / -apply-pihole / -reconcile	xenon	RabbitMQ sor.events	CDC pipeline — F-08 gap flag
BIND (akoria.net)	xenon (primary); radium, steel (secondary)	:53	+ RPZ blocklist + ~100 split-horizon "bare" zones OBS
Samba AD DC + Keycloak	radium	LDAP/Kerberos/OIDC; sso.akoria.net	authoritative akoria.org OBS
Forgejo	cesium	git.akoria.net	STATIC_ROOT_PATH on internal storage SoT

SERVICE	HOST	ENDPOINT	NOTES / FLAG
MariaDB — SoR/CMDB	cesium (primary), benzene (mirror)	db.akoria.net → benzene	separate DB from monitoring SoT
RabbitMQ	benzene	amqp://...@10.5.2.50:5672/ solari	notify.events + sor.events; no canonical broker config in repo flag
step-ca internal CA	chlorine	ca/acme/pki.akoria.net	X.509 + S/MIME + ACME + SSH CA SoT
Fleet netboot HTTP	benzene	:8080	staged, inert — contained (F-07) flag

6.3 Ports & sockets

7701 / 7702 / 7703	SCP over tls+tcp (nng + mbedTLS mTLS) — client, monitor, and control channels into solariServer.
/run/solari/solariCtl.sock	AF_UNIX line protocol — the only mutation path from the dashboard to the C core.
9443 / 443	Dashboard HTTPS (intranet additive vhost / public subdomain vhost).
53	DNS — BIND on xenon/radium/steel + Pi-hole on helium/mercury.
5672	RabbitMQ AMQP on benzene (vhost solari).

Data model & System of Record

Two MariaDB databases, deliberately separate: a monitoring datastore that records what the fabric observes, and a System of Record that holds what the fabric is supposed to be. Discovery flows from the first into the second; DNS and Pi-hole flow out of the second.

7.1 Two databases

- Monitoring DB** ~30 InnoDB tables under db/migrations/001-019 — nodes, reports, probe results, alerts and rules, topology edges, discovery, panel/criticality, lifecycle tombstones. All timestamps stored UTC.
- SoR / CMDB** Separate DB on cesium, mirrored to benzene. The canonical entity spine is defined in netdb/sor/schema.sql (netdb/sor/migrations/ holds only incremental deltas): entities, hardware_models, hardware_units, locations, networks, subnets, vlans, interfaces, ip_addresses. Change-data-capture rides a separate sor_outbox table defined in deploy/sorsync/sql/01-outbox.sql.

KNOWN GAP — F-12, SCHEMA LIFECYCLE

The monitoring schema is **not reproducible from a clean checkout** as built: schema.sql is stale (reflects only migration 001), there is a **duplicate ordinal 016** (two files, currently inert), and there is no single migrator with immutable IDs, checksums, and an applied-ledger. Phase-2 (I2) adopts one migrator and resolves the duplicate. Until then, the live database — not the tree — is the truth of what columns exist.

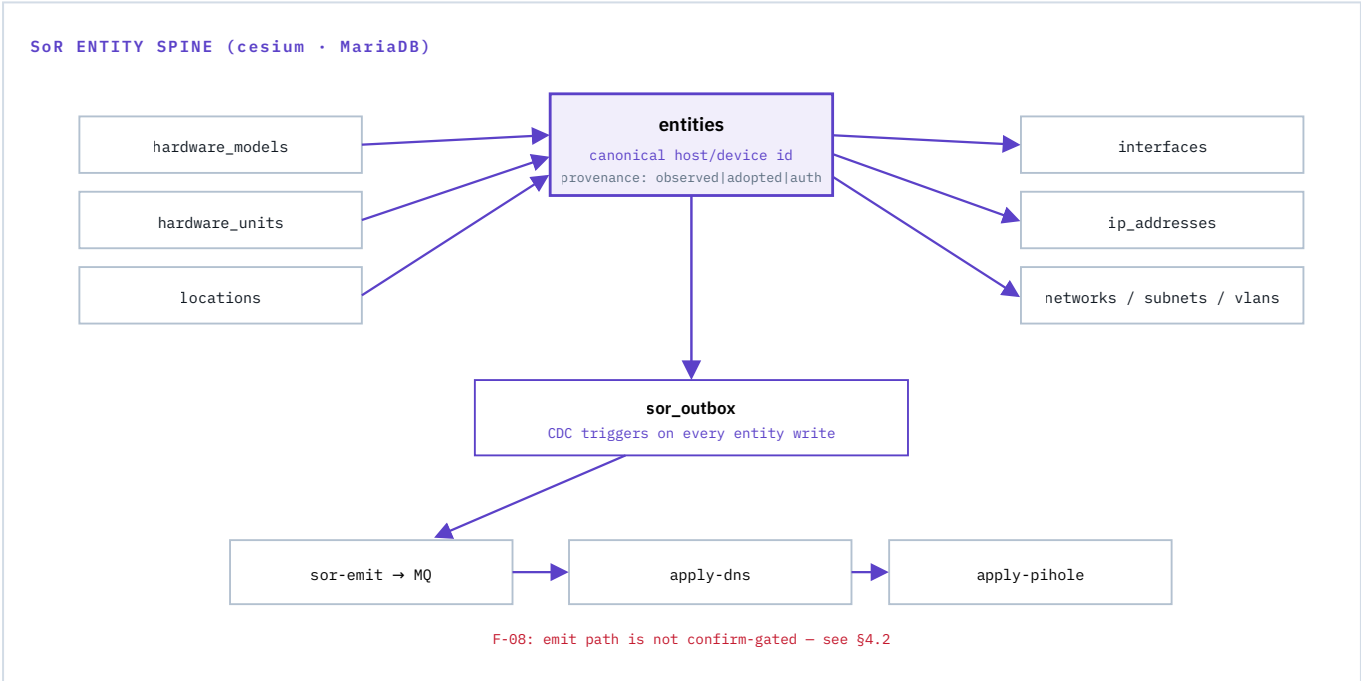


Figure 7.1 — The SoR entity spine and its CDC outbox. Every page in the planned integration (I2/I4) hangs off entities. **OBSERVED** from netdb/sor/schema.sql + deploy/sorsync/sql/.

7.2 Provenance as a first-class column

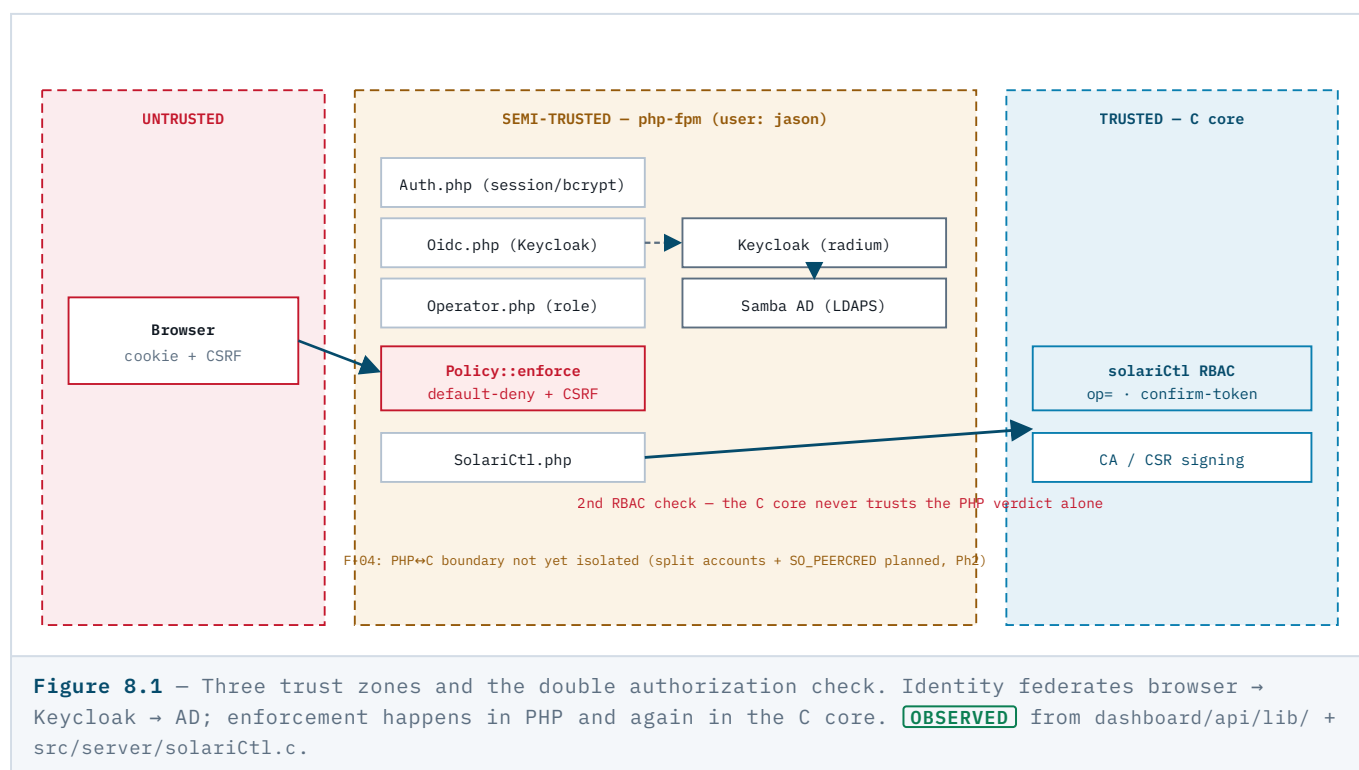
The SoR distinguishes how it learned each fact: observed (seen by discovery), adopted (an operator promoted an observation to intended state), and authoritative (declared, the source of truth others reconcile to). The planned I2 track surfaces this provenance on every entity page, so the dashboard can never again present a discovered guess as a declared fact — the same honesty discipline that governs this document's **OBS**/**INF** flags.

Identity, authorization & trust boundaries

Authorization is enforced twice on the dangerous path — once in PHP before a command is formed, and again in C before it executes — and identity is derived from certificates on the wire, not from claims in a payload.

8.1 The human auth chain

A browser request is authenticated and authorized through four cooperating libraries before any handler runs. Local accounts use bcrypt; federated login uses Keycloak's OIDC authorization-code flow (confidential client), with Keycloak in turn federated to Samba AD over LDAPS. The legacy direct-LDAP path is a **fail-closed stub** — it denies rather than degrading open.



8.2 Machine identity on the wire

Machines do not authenticate with claims. Every SCP peer presents a client certificate whose CN encodes `<role>.<fqdn>`; the ingest path derives the role from the verified certificate and rejects any message type illegal for that role (`ERR_AUTH_ROLE`). The planned F-01 work makes the binding total — cert SAN/CN ↔ node ID ↔ role ↔ FQDN as one canonical identity contract, so a valid certificate can never be replayed under a different node's identity.

8.3 Trust-boundary summary

WHERE TRUST CHANGES HANDS – AND THE FINDING THAT HARDENS EACH EDGE.

BOUNDARY	CONTROL TODAY	HARDENING	SEV
Browser → PHP	session + default-deny Policy + CSRF/Ori- gin	claims validation, break-glass disable, rate-limit, role revalidation (F-14)	P1
PHP → C (so- lariCtl)	local UNIX socket; op= attribution; con- firm-token for destructive verbs	framed I/O + deadlines + caps (F-11); split least-priv accounts + SO_PEERCREC (F-04)	P0
Node → server (SCP)	mTLS; role-from-cert-CN; illegal-type re- ject	frame↔cert identity binding (F-01); mTLS invariant not configurable (F-29)	P0
Server → DNS/ AD	SoR-driven render + apply	render+validate+atomic-switch+rollback (F-17); AD/ LDAP write paths frozen	P1

Developer workflow & build

One CMake tree builds the C tier; the dashboard has no build at all. The house rules are strict and load-bearing — they are why the system can be reasoned about.

9.1 Building the C tier

The control plane is ISO C99 built with CMake, wrapped by a thin top-level Makefile for convenience. Optional subsystems compile behind feature flags, so a minimal client can be built without the SQLite spool or the nng transport.

```
# from repo root — the Makefile just drives CMake
make          # configure + build all binaries into build/
make test     # run the C test suite (golden SCP frames, unit)

# feature flags (CMake cache):
SOLARI_WITH_IO=ON      # nng transport + reporter (solarinet.c)
SOLARI_WITH_SQLITE=ON  # client-side durable spool (solarispool.c)
```

Wire compatibility is protected by **golden-frame fixtures**: canonical SCP frames are committed per protocol version and are immutable — a change that alters the bytes of an existing version fails the test rather than silently breaking older peers.

9.2 The dashboard has no build step

HARD HOUSE RULE

There is **no bundler, no package.json, no composer.json, and no Node.js** anywhere in the runtime. **OBSERVED** — none of those files exists in the tree. React and Babel are vendored under dashboard/public/; JSX in screens*.jsx is transpiled in the browser at load. To develop the UI you edit a .jsx file and reload — there is nothing to compile. Introducing a build step into this code is forbidden by project policy, not merely discouraged.

9.3 Coding standards

- Language** Standard C (K&R/ANSI) for the application; functional PHP 8+ for the dashboard; browser-only JS. **No server-side JavaScript, ever.**
- Naming** camelCase is the house standard for variables, functions, and named structures across C and PHP alike — including where the language community would use snake_case.
- Paradigm** Prefer functions to objects. Classes must earn their place with real shared-state invariants; the C core uses an opaque-handle pattern, not OOP emulation.
- Comments** Every program, function, and non-obvious control structure carries a brief purpose / input / output statement. Explain intent and edge cases, not the obvious.

9.4 Tests & the CI gates being added

Dashboard authorization has PHP tests (tests/dashboard/test_authz_enforce.php, test_authz_dispatch.php) exercising the default-deny matrix and the request-method-to-verb dispatch. There is no phpunit or CONTRIBUTING in the tree yet. The stabilization program adds a battery of CI gates: a live-DB gate (SOLARI_TEST_DB=1), TLS-loopback + cert-identity failure tests, a generated route×method×role matrix with CSRF/Origin tests, submodule init before dashboard tests, migration clean-install / upgrade / idempotent-rerun checks, shellcheck/ASAN/UBSAN/fuzzers, and fault injection (broker loss, lease race, partial I/O, backup restore). A **cross-lab review gate** is required before committing to Phase-2 architectural choices — the original review was single-lab.

Operations & deployment

SolariNet is deployed as hardened systemd units driven from the `deploy/` tree. This section covers how it runs, and — just as important — which capabilities are deliberately switched off.

10.1 Deployment layout

<code>deploy/host-configs/</code>	Per-host systemd units and Apache vhosts (e.g. <code>xenon/solarinet-server.service</code> , <code>apache-solarinet.conf</code>). Units are hardened (namespacing, restricted syscalls).
<code>deploy/fleet/</code>	Ansible inventory + a netboot <code>catalog.json</code> . The netboot HTTP service on <code>benzene:8080</code> is staged and inert — see freeze below.
<code>deploy/dns-bare/</code> , <code>netdb/</code>	Zone generation from the SoT: <code>gen-zones.py</code> renders <code>akoria.net</code> ; <code>sync-net-to-org.py</code> mirrors into the AD realm hourly; <code>gen-bare.py</code> produces the ~100 split-horizon per-host zones.
<code>deploy/sorsync/</code>	The CDC units — <code>sor-emit</code> , <code>sor-apply-dns</code> , <code>sor-apply-pihole</code> , <code>sor-reconcile</code> — running on <code>xenon</code> .

10.2 Frozen capabilities

FEATURE FREEZE — PHASE 0 CONTAINMENT

The following are implemented but held **LAB**-only until their Phase-blockers close. They must not be enabled in production, and no new features may be built until Phase 4 gates pass.

- **Tab5 approvals** — approval display not cryptographically bound to the approval (F-06/F-24).
- **One-button PXE provisioning** — executes unauthenticated artifacts as root; key material left staged (F-07/F-16).
- **Arbitrary MCP SQL** — replaced with fixed tools/views + a read-only account (F-18).
- **Opie automation** — "read-only" model is not confidentiality-safe; sandboxed (F-15).
- **Remote CA automation** — kept manual on `chlorine`.
- **LDAP/AD write paths** beyond current use — frozen; AD is read-federated only.

10.3 Operational cautions

BEFORE YOU ACT

- **Alarm tests are loud.** The status-panel and alert paths drive real physical notification (LED panel on lithium, push bridges). A test alarm rings the actual fabric — announce it first.
- **Destructive verbs need attribution.** DECOMMISSION, RETIRE, SIGN, DEPLOY, FLEET_*, and the *_REMOVE/_DEL verbs require op=; DECOMMISSION needs the two-step confirm-token.
- **Backups are not proven restorable.** F-27 — nonempty ≠ restorable; a restore drill is Phase-3 work. Do not assume a backup will roll back a bad change.
- **DNS/AD apply has no rollback yet.** F-17 — apply paths lack atomic switch + rollback semantics; a bad zone render is not automatically reversible today.

10.4 Health philosophy

The system's guiding operational rule, learned from the 2026-07-07 cesium incident (Forgejo served 500s for ~26 hours while a bare TCP check stayed green): **a socket accepting a connection is not health**. The host-health contract requires application-level liveness, and the planned freshness envelope {lastSuccess, ageSeconds, state} (I3) makes every page state its own staleness rather than presenting cached data as live.

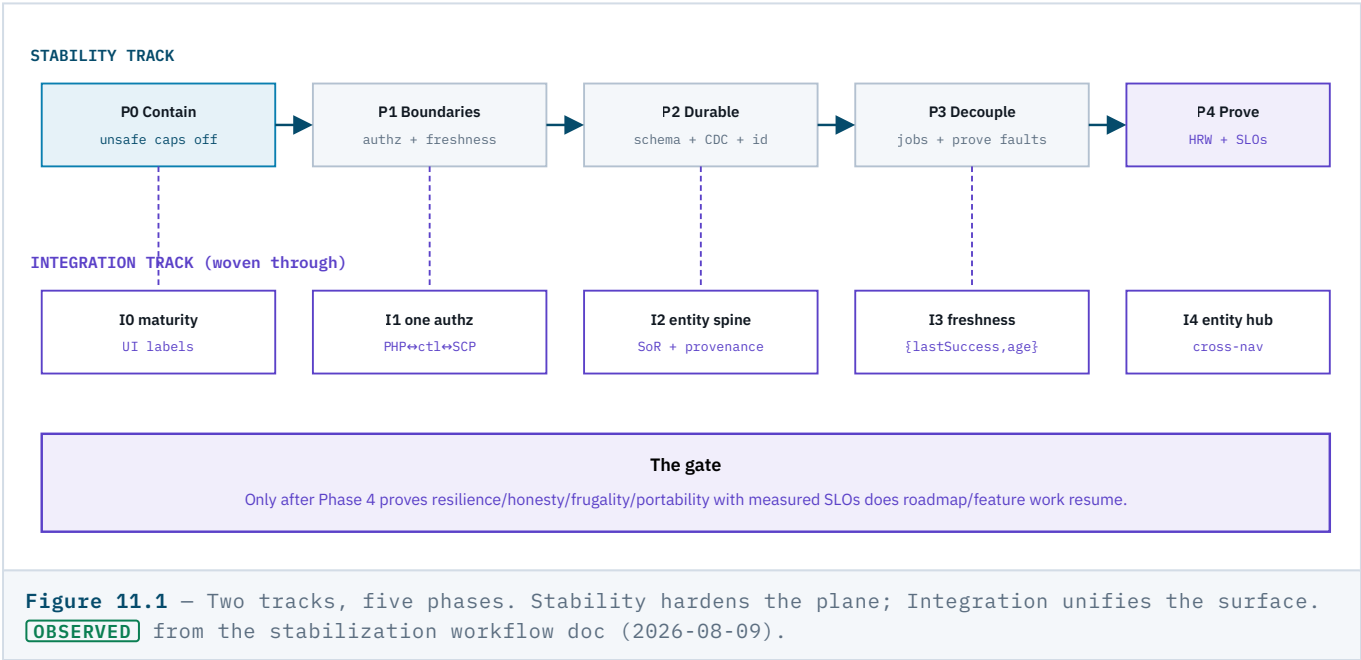
Planned architecture & the stabilization program

An independent architecture review found that feature breadth had outrun enforcement and durability. The response is a declared feature freeze and a phased program with two goals: make the system honest and make it whole. Nothing new ships until it is done.

11.1 Two goals

- Stability** "Done" means authorized and durably persisted and correctly delivered and actually applied and currently fresh — not merely "a socket accepted a message."
- Integration** One entity spine, one freshness/health contract everywhere, one authorization matrix, consistent cross-navigation — pages and services tightly interlinked rather than parallel silos.

11.2 Phase roadmap



11.3 What is real vs. designed today

CAPABILITY MATURITY – THE DISTINCTION THIS DOCUMENT EXISTS TO PRESERVE.

CAPABILITY	MATURITY	NOTE
Client reporting + ingest + MariaDB persis- tence	PRODUCTION	load-bearing across the fleet
Dashboard authz (default-deny + CSRF)	PRODUCTION	Phase-0 hardening landed (2d74adf/1812a9e)
SoR + CDC → DNS/Pi-hole	LAB	works; can lose events (F-08) until Phase-2

CAPABILITY	MATURITY	NOTE
Monitor tier + HRW k-of-n redundancy	PROTOTYPE	coded, not load-bearing; nearly all nodes are client-role (F-19/F-20)
Two-server HA / lease failover	PROTOTYPE	production is fleet-of-one; prove in Phase-3/4 (F-09)
Fleet PXE provisioning	LAB	frozen — unauthenticated-as-root (F-07)
Decommission lifecycle	LAB	script exists; not first-class/dashboard-driven yet

11.4 Findings register — F-01 ... F-31

The complete register from the 2026-08-09 review, with severity and program disposition. This is the authoritative backlog; each item is Adopted (fix as designed), Adapted (fix to the real fleet-of-one), Contained (kept lab-only now, fixed later), or Folded (merged into another track).

FINDINGS F-01..F-31 — FINDINGS & RECOMMENDATIONS 2026-08-09 · DISPOSITIONED IN THE STABILIZATION WORKFLOW

ID	TITLE	SEV	DISPOSITION
F-01	Node certs not bound to frame identity	P0	Adopt · I1 / Ph2
F-02	Dashboard authorization systemically incomplete	P0	Adopt (default-deny) · I1 / Ph1 ✓
F-03	Cookie-auth control requests have no CSRF defense	P0	Adopt (token+Origin) · Ph1 ✓
F-04	PHP/C trust boundary not isolated	P0	Adopt (split accounts, SO_PEERCREDS) · Ph2
F-05	Viewer-triggerable scan can starve the lease	P0	Adopt · gate Ph1, job-runner Ph3
F-06	Tab5 approval not cryptographically bound	P0	Contain now · Ph0
F-07	PXE runs unauthenticated artifacts as root	P0	Contain now, signed-manifest Ph3
F-08	SoR CDC can permanently lose events	P1	Adopt (confirms+reconcile) · I2 / Ph2
F-09	Server HA semantics internally inconsistent	P1	Adapt (fleet-of-one) · prove Ph3/4
F-10	Health & convergence records can lie	P1	Adopt (persist once, fsync) · I3 / Ph2
F-11	solariCtl wire protocol unsafe both ends	P1	Adopt (framed I/O+caps) · Ph1
F-12	Schema lifecycle not reproducible	P1	Adopt (one migrator, ledger) · I2 / Ph2
F-13	Browser freshness violates honesty goal	P1	Adopt — the hinge · I3 / Ph1
F-14	Auth/session controls incomplete	P1	Adopt (claims, rate-limit) · Ph1
F-15	Opie "read-only" not confidentiality-safe	P1	Contain now, harden Ph3
F-16	Fleet provisioning leaves key material staged	P1	Adopt (node keys, one-time CSR) · Ph3
F-17	DNS/AD apply lacks safe deploy semantics	P1	Adopt (atomic+rollback) · I2 / Ph2–3
F-18	Arbitrary MCP SQL not safely constrained	P1	Contain (fixed views, RO acct) · Ph0
F-19	Monitor probing sequential/capped	P2	Adopt as batch · Ph3–4

ID	TITLE	SEV	DISPOSITION
F-20	HRW membership unproven; claim too broad	P2	Adopt + doc-fix now · Ph3–4
F-21	Spool has no quota/age; sequence reuse	P2	Adopt as batch · Ph3–4
F-22	SoR multi-step writes untransacted	P2	Adopt · folded into Ph2 (F-17)
F-23	Decommission sends before durable confirm	P2	Adopt as batch · Ph3–4
F-24	Tab5 secret-at-rest unproven	P2	Adopt as batch · Ph3–4
F-25	UniFi false-green	P2	Fold into I3 · pulled into Ph1
F-26	Authbroker fails open on empty token	P2	Adopt as batch · Ph3–4
F-27	Backups nonempty ≠ restorable	P2	Adopt as batch · Ph3–4
F-28	Sparse referential integrity / orphans	P2	Adopt as batch · Ph3–4
F-29	mTLS configurable, not invariant	P2	Adopt as batch · Ph3–4
F-30	Warning-clean CI + sanitizers + fuzzing	P3	Adopt into CI gate · ongoing
F-31	CSP/HSTS/headers under no-build constraint	P3	Adopt into CI gate · ongoing

✓ = landed on the current branch (stabilize/phase-0-containment) as of this revision. All other items are open per their phase.

Appendices

A Glossary

SCP	SolariNet Control Protocol — length-prefixed 32-byte big-endian header + TLV payload + CRC-32, over nng + mbedTLS mTLS.
HRW	Highest-Random-Weight (rendezvous) hashing — coordinator-free assignment of probe targets to monitors.
SoR / CMDB	System of Record / Configuration Management Database — the declared-intent datastore on cesium.
CDC	Change Data Capture — the outbox-and-consume path that propagates SoR changes to DNS/Pi-hole.
solariCtl	The in-process operator bridge inside solariServer; AF_UNIX line protocol; holds CA signing material.
RPZ	Response Policy Zone — the DNS blocklist zone served by BIND on xenon.
F-nn	A numbered finding from the 2026-08-09 architecture review (see §11.4).

B Host quick-index

EVERY ELEMENT NAME → ADDRESS → ONE-LINE ROLE (SEE §5 FOR FULL DETAIL).

HOST	ADDRESS	ROLE	HOST	ADDRESS	ROLE
chemistry	10.0.0.1	gateway (UDR7)	radium	10.1.0.10	AD DC + Keycloak
laboratory	10.0.0.3	core switch	cesium	10.1.0.200	Forgejo + SoR
xenon	10.0.0.20	SolariNet server + DNS	amino	10.1.0.100	n8n
carbon	10.0.0.10	NAS (DXP4800 Plus)	benzene	10.5.2.50	MariaDB + netboot
beaker / flask	10.0.0.7/.8	WiFi 7 APs	nitrogen	10.5.2.20	dev
helium / mercury	10.0.0.254/.253	Pi-hole pair	quanta	10.6.0.10	Grafana
steel	10.0.0.11	DNS secondary	valence	10.6.0.20	monitor outpost
silicon	10.0.0.250	edge proxy (Caddy)	chlorine	10.7.0.10	internal CA
silver	10.0.0.85	printer	photon	10.7.0.1	NVR hub
hydrogen	10.0.1.50	workstation (Mac)	tachyon	10.6.6.10	GPS/PTP time

C Methodology & provenance

This reference was assembled from three cross-checked sources, all captured on 2026-09-17: the committed source tree on branch stabilize/phase-0-containment; the rendered live configuration on xenon (BIND zones, systemd units, route and ARP tables, read via `sudo cat` and read-only SSH); and an authorized live discovery pass. Discovery method was `nmap -sn -PR` across the 16 zoned /24s plus read-only SSH fact-gathering; the offline IEEE OUI table resolved MAC vendors. A full service/OS scan and SNMP polling were **not** completed — those columns are blank by honest omission, and no MAC, model, or address in this document was synthesized to fill a gap. Where a fact could not be observed, it is marked **INFERRED** or left as an explicit gap in §5.3.

LIVING DOCUMENT

The inventory is a point-in-time snapshot; hosts move and the SoR drifts. The reconciliation queue in §5.3 (five unnamed hosts, the lithium subnet, the iron/steel rename) is the current delta between the committed record and observed reality. Re-run the discovery pass and this build to refresh.

SolariNet Developer & Architecture Reference · Revision 1.0 · 2026-09-17 · Internal · github.com/soren42/Solarinet · Set in IBM Plex Sans & Mono · Rendered with WeasyPrint from self-contained HTML — no build step.